

Explainable AI-Based Personalized Learning Path Recommendation System for University Students

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Abstract—As digital learning platforms have become increasingly common in higher education, they have generated huge amounts of data on student interactions and performance that enable personalised learning route recommendations. But present educational recommender systems usually provide little transparency, cannot accurately simulate the time-varying forgetfulness and knowledge of students, and are rarely evaluated in a way that establishes which of their components actually contribute. This paper proposes an Explainable Artificial Intelligence (XAI)-based personalized learning path recommendation system consisting of a knowledge-graph-enhanced Deep Knowledge Tracing module, a prerequisite-constrained planner, and a priority-backtracking explanation module. Experiments on the Open University Learning Analytics Dataset (OULAD), under five-fold cross-validation with folds grouped by student, give an AUC-ROC of 0.9538 ± 0.0019 and an RMSE of 0.2621 ± 0.0026 , exceeding DKT (0.9435 , $p = 0.0009$) and a GNN baseline (0.9502 , $p = 0.022$) after Holm correction. A central finding concerns evaluation rather than architecture. Removing knowledge-graph propagation changes accuracy by -0.0001 ($p = 0.77$), which would ordinarily be read as the graph contributing nothing; that reading is an artefact of the measurement, because accuracy is scored only at assessment positions and only 89 of 237 concepts carry assessments. Under a cold-concept protocol that withholds all assessments for a subset of concepts and scores only those, the same graph improves accuracy by $+0.0325$ ($p = 0.0003$, Cohen’s $d = 5.06$), winning on every fold. The planner produces zero prerequisite violations across 906 decisions evaluated against the concepts learners actually studied next, and explanation quality is assessed by objective fidelity, sufficiency and stability rather than by subjective rating. All code and results are released publicly.

Index Terms—Explainable AI (XAI), personalized learning path, knowledge tracing, knowledge graphs, educational data mining, ablation methodology, OULAD dataset.

I. Introduction

Higher education’s quick embrace of digital learning systems has opened up new possibilities for individualized, data-driven instruction. Yet despite differences in prior knowledge, learning speed and career aspirations, the

learning process remains the same for all students in conventional courses. In complex domains such “one size fits all” approaches cause course mis-selection, knowledge gaps and less effective learning [1]– [3]. Personalized learning path recommendation systems can overcome these constraints by evaluating student interactions to determine appropriate activities, but most systems in use concentrate on prediction accuracy and offer little explanation for any given recommendation [4].

Designing a system that describes a learner’s evolving knowledge state, sequences activities optimally, and explains its choices is therefore a major challenge. Most knowledge tracing methods assess learning rather than optimize progression, while collaborative filtering suffers from sparsity and weak personalization [5], [6]. Automated sequencing can optimize for long-term outcomes [7] [8], but its decisions remain opaque to students and advisers, and the connection between knowledge graphs, sequential decision-making and temporal knowledge modeling is still underexplored in education [10], [11].

A further limitation motivates much of this work. Such systems combine several components — a sequence model, a structural prior, a planner — each justified by an ablation showing that its removal degrades accuracy. We find this evaluation can be systematically uninformative: when a structural prior exists precisely to support concepts whose supervision is scarce, and accuracy is measured only where supervision is abundant, the ablation tests the component in the one regime where it cannot matter.

The main contributions of this work are as follows.

- 1) A knowledge-graph-enhanced knowledge tracing model with a learned per-head temporal decay, attaining AUC-ROC 0.9538 ± 0.0019 and calibration error 0.0054 on OULAD under student-grouped five-fold cross-validation.
- 2) Evidence that conventional ablation cannot detect

the contribution of a knowledge graph, together with a cold-concept protocol that can. Prerequisite propagation changes standard accuracy by -0.0001 ($p = 0.77$) yet improves accuracy on concepts lacking direct supervision by $+0.0325$ ($p = 0.0003$, $d = 5.06$).

- 3) A prerequisite-constrained planner whose pedagogical guarantee is enforced structurally rather than learned, verified at zero violations across 906 recommendation decisions evaluated against observed learner behaviour.
- 4) Three objective measures of explanation quality — fidelity, sufficiency and stability — in place of subjective rating. One revealed that explanations accounted for only 0.2% of the benefit they described, a defect no rating scale would surface.
- 5) A reproducible pipeline from raw data to every reported figure, released publicly.

II. Literature Review

Educational recommender systems have progressed from one-size-fits-all delivery to adaptive personalization. Early content-based and collaborative filtering approaches suffered cold-start problems, data sparsity and limited scalability [1] [2]. Hybrid models improved accuracy and robustness [3] but forecast learner preferences rather than modeling knowledge acquisition or optimizing sequential pathways [4].

Deep Knowledge Tracing (DKT) [5] simulates the latent knowledge state of learners over time. Unlike Bayesian Knowledge Tracing, which uses static parameters, DKT learns temporal dependencies directly from data [6], and attention-based successors [7], [21] attend directly to any earlier interaction rather than propagating through a recurrent chain [9]. Reinforcement Learning enhances path recommendation by formalizing the process as a Markov Decision Process and maximizing long-term learning rather than engagement; combined with knowledge tracing it has been reported to reduce dropout and completion time [8], [18]. Knowledge graphs offer a complementary route to structured, explainable recommendation by explicitly modelling concepts, course units and their interactions [10], [15], with Bloom’s taxonomy grounding path design [12] and graph neural networks handling the resulting relational structure. Multi-objective frameworks have optimized dropout risk, satisfaction and completion [14] [13], and neuro-symbolic and generative approaches have begun to explore explanation generation [15], [16].

Four gaps remain. First, explainability is added as an afterthought rather than being integral to recommendation optimization [17]; SHAP-based explanations dominate in education and are applied mainly at the global level [18], and where explanations are evaluated at all it is usually by participant rating, which measures whether readers found an explanation agreeable rather than whether it is faithful to the model. Second, many frameworks ignore

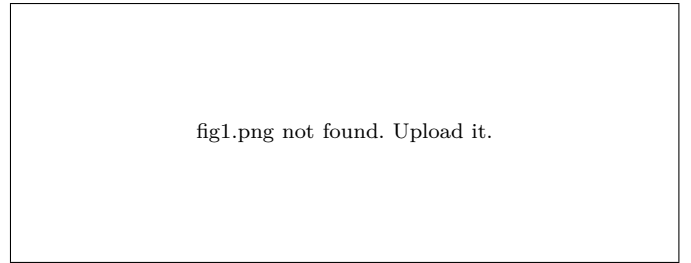


Fig. 1. Overall XAI-based personalized learning recommendation framework

the dynamic evolution of knowledge states, including forgetting, compromising the pedagogical integrity of suggested trajectories [19]. Third, applying knowledge graphs to sequential decision-making and explanation remains at an early stage [20]. Fourth, and most relevant here, the contribution of such structural priors is customarily established by an ablation on aggregate accuracy, without examining whether that metric can observe the component at all.

A. Research Gap

Existing educational recommender systems focus on prediction and preference-based suggestion, rarely integrating knowledge graphs, temporal knowledge tracing, constrained path optimization and pedagogically meaningful explanation. This limits customization, transparency and effective learning-path optimization. A further gap concerns validation: component contributions are reported on metrics that may be structurally unable to detect them, and explanation quality is reported by opinion rather than measurement.

III. Methodology

A. Overall Framework

The proposed system combines Knowledge Graph Modeling, Explainable AI (XAI), prerequisite-constrained path optimization and Deep Knowledge Tracing to recommend personalized learning trajectories, through learner knowledge estimation, sequential path optimization and explanation production [5]. Student interaction history is first processed to estimate the learner’s changing knowledge state; prerequisite and co-requisite relationships in the knowledge graph then constrain a planner that selects the next learning task; finally each recommendation receives a pedagogical explanation from a priority-backtracking system [15].

B. Data Acquisition and Pre-processing

Evaluation uses the Open University Learning Analytics Dataset (OULAD) [3], covering clicks, enrollments, assessment results and student attributes. Interaction records are ordered chronologically, categorical features encoded and numerical attributes standardized. OULAD timestamps are day-granular with no intra-day resolution,

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Fig. 2. Knowledge graph enhanced knowledge tracing

so a session is one learner-day rather than an inactivity threshold. Temporal features capture session volume, elapsed time since the previous interaction, and mastery change. The processed data is split 80:10:10.

OULAD supplies no concept annotations, prerequisite relations or material titles, so the concept graph is derived rather than read off. A concept is a (module, week-of-study) pair: the material a module presents in a given week. Week placement uses the week_from field where present (17.6% of materials) and otherwise the median access day; on the 1,121 materials carrying both, the two agree at Spearman $\rho = 0.975$. Edges come from three rules: sequence edges between consecutive weeks, assessment-coverage edges from the four preceding weeks to an assessment’s week, and edges mined from behaviour. A mined edge between concepts a and b is accepted only where behaviour shows both consistent ordering and a measurable effect on later success,

$$\begin{aligned} \text{prec}(a, b) &= \frac{|\{u : \tau_u(a) < \tau_u(b)\}|}{|\{u : a, b \in h_u\}|}, \\ \text{lift}(a, b) &= \frac{P(y_b | a \prec b)}{P(y_b)}, \end{aligned} \quad (1)$$

where $\tau_u(\cdot)$ is learner u ’s first interaction with a concept, requiring support above 100 learners, $\text{prec} \geq 0.75$ and $\text{lift} > 1.1$. Requiring lift matters: precedence alone admits pairs separated only by the academic calendar, about half the candidates. The resulting graph is acyclic without intervention.

C. Knowledge Graph-Enhanced Knowledge Tracing

Student knowledge concepts are estimated through past encounters by adopting a knowledge tracing model that is based on a Transformer [9]. Temporal attention depicts the dependency of learning activities and changes in learners’ knowledge over time. A Graph Convolutional Network (GCN) [10] simultaneously encodes the precondition relationships in the knowledge graph, producing a concept embedding that is recomputed at every forward pass so that gradients propagate into the graph structure. The combined temporal learner representation and graph-based concept representation make up the current learner state. The system may take into account the learner’s past progress as well as the structural links between ideas thanks to this integrated representation.

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Fig. 3. Prerequisite-constrained learning path optimization

Forgetting is modelled explicitly, as an additive attention bias that decays with elapsed time,

$$b_{ij}^{(h)} = -\text{softplus}(\gamma_h) \log(1 + \Delta_{ij}), \quad (2)$$

where Δ_{ij} is the number of days between interactions i and j and γ_h is learned separately for each attention head, so that one head may attend to recent activity while another retains longer memory. Setting $\gamma_h = 0$ recovers ordinary attention and forms the temporal ablation reported in Section V.

Two signals drive the model. Clickstream interactions update the hidden state but contribute no training loss; assessment interactions are the supervised positions, and binary cross-entropy is computed over those alone. This dual-signal construction is what makes a sequence model of length 200 trainable on a dataset in which learners average 8.7 assessments each. Because predicted mastery is displayed to learners inside explanations, the output probabilities are calibrated by temperature scaling [22], with the temperature fitted on one half of each fold’s held-out predictions and evaluated on the other.

D. Prerequisite-Constrained Path Optimization

The recommendation problem is formulated as a Markov Decision Process, where each course, module or activity is a course of action, and the learner state is the knowledge and learning context of the learner. The state concatenates the estimated mastery vector over all concepts, a coverage mask of concepts already recommended, the static learner features and the remaining budget. The reward combines learning progress, prerequisite satisfaction, a redundancy penalty and a small step cost.

Rather than expecting a learned policy to discover the pedagogical constraint, it is encoded directly in the set of admissible actions: a concept is admissible only if the learner has not already mastered it, every prerequisite is at or above a readiness threshold, and it belongs to an enrolled module. An unprepared recommendation is therefore impossible by construction rather than merely unlikely, as Section V verifies.

The policy evaluated here is greedy with respect to one-step predicted improvement: each admissible concept is simulated as studied, the resulting mastery vector is compared against a baseline in which the same period elapses without study, and the concept giving the largest

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Fig. 4. Explainable AI module for recommendation justification

total improvement in log-odds is recommended. Two details are load-bearing. The baseline must be an idle period rather than the present state, because studying advances the clock and the decay of (2) then attenuates every earlier interaction, so measuring against the present moment makes every action appear harmful. The comparison must be in log-odds, because engaged learners saturate near 0.98 where probability differences round to zero and the ranking degenerates. Mastery is bimodal, so the mastery and readiness thresholds are the 70th and 30th percentiles of each learner’s own distribution rather than fixed values. A learned policy over the same state and reward is left to future work; Section V reports the headroom for one.

E. Explainable Recommendation and Evaluation

Once an activity has been chosen, the priority-backtracking interpreter follows its choice across the knowledge graph to find pertinent mastery levels, prerequisite links, and knowledge gaps. Each prerequisite ancestor is scored by its shortfall below the readiness threshold, discounted by graph distance, so that an unmet prerequisite immediately preceding the recommendation outranks an equal shortfall further back; ancestors are reported in descending order of that score. Counterfactual statements are computed by raising a gap concept’s mastery to the mastery threshold and re-evaluating the planner’s score, so the claimed effect of closing a gap is measured rather than asserted. These elements are then turned into pedagogically meaningful explanations for students and academic advisors [15].

AUC-ROC, RMSE and expected calibration error are used to assess knowledge-tracing performance. Learning-path quality is assessed against observed behaviour rather than simulation: each held-out learner’s history is truncated at several points, each planner is asked for a recommendation, and the ground truth is the set of concepts that learner actually engaged with over the following interactions, with only learners who passed or achieved distinction contributing ground truth. Explanation quality is assessed by three objective measures — fidelity, sufficiency and stability — defined below, rather than by participant rating. Baselines comprise a majority-class predictor, SAKT, DKT and a GNN-based recommender. Ablation experiments are performed to remove certain elements from the proposed framework to

determine the effect on overall performance, under both the conventional protocol and the cold-concept protocol introduced in Section V.

IV. Implementation

A. Development Tools and Computing Environment

The framework is implemented in Python 3.11. The Transformer-based knowledge tracing model and the graph convolution component use PyTorch. Data preparation is performed in SQL against an embedded DuckDB database, so every transformation from raw records to model input is auditable without reading application code. NetworkX builds and traverses the knowledge graph, and the mlxtend implementation of FP-Growth mines co-requisites. Training used a single NVIDIA T4 GPU; all data processing, graph construction, planning and inference run on CPU, on an Intel Core i7 workstation with 16 GB of RAM. Data preparation takes about 2.5 minutes and training one configuration about 25 minutes.

B. Dataset Configuration and Preprocessing

The main dataset for implementation and assessment is the Open University Learning Analytics Dataset (OULAD). The data set includes clickstream interaction data, student enrollment, assessment scores and student demographics, comprising 32,593 enrolments, 10,655,280 clickstream events and 173,912 assessment records across seven modules. A pipeline built on SQL, Pandas and NumPy preprocesses the raw records into sequential learning data. Interactions are presented in chronological order and grouped into learner-days. The categorical variables such as gender and education level are converted to one-hot variables and the missing assessment scores are handled using median imputation, applied only where a submission exists.

Two properties of the data required specific treatment and are reported here because both materially affect what the reported metrics mean.

Label construction. OULAD records only submitted assessments. Binarising at the pass mark of 40 yields a 95.6% positive rate, the fifth percentile of submitted scores being the pass mark itself: the failures are absent from the records rather than from reality. Of 323,925 expected submissions, 150,013 never occurred. Treating non-submission as failure, restricted to assessments falling while a learner remained registered, gives a 67.9/32.1 balance and makes the target successful completion.

Concept supervision is sparse and uneven. The implementation contains 237 concept nodes, 724 prerequisite relationships and 302 co-requisite relationships, with no cycles requiring removal. Assessments exist for only 89 of the 237 concepts; mastery estimates for the remaining 148 are reached through prerequisite propagation. This is the condition that Section V examines directly. Temporal information such as the volume of study activity, elapsed time since the previous interaction and progression in

mastery of concepts is added to represent changes in learner behaviour and knowledge over time.

C. Model Implementation Specifications

A Transformer-based knowledge tracing encoder with two hidden layers, a hidden dimension of 128, four attention heads, and a maximum sequence length of 200 interactions is used in the learner modeling component. The three graph convolution layers in the graph convolution (GCN) component are followed by layer normalization and ReLU activation. 64-dimensional concept embeddings are produced by the resulting graph encoder. These embeddings are combined with the temporal learner representation to form the state on which the planner operates. Dropout is 0.2. Optimisation uses Adam at learning rate 10^{-3} with cosine annealing, weight decay 10^{-4} , batch size 64, and at most 12 epochs with early stopping at patience 3. The complete model has 466,569 parameters. Random seeds are fixed and recorded alongside every result.

The planner evaluates all admissible concepts in one batched forward pass, which keeps recommendation latency at roughly 160 ms on CPU. The priority-backtracking interpreter keeps track of the relationship between a suggested idea and the concepts that it requires by means of recursive graph traversal to a maximum depth of four. It draws out pedagogically meaningful explanations for each recommendation, based on the learner proficiency levels and relevant gaps in knowledge.

D. Evaluation and Validation Implementation

To check the implemented system against the majority-class, SAKT, DKT and GNN baselines, a consistent evaluation technique is used. To evaluate predictive performance, AUC-ROC, RMSE and expected calibration error are employed, while learning-path quality is measured by NDCG@5, Hit@5, Precision@5, prerequisite violation rate and concept coverage against the concepts learners actually studied next. Explanation quality is measured by fidelity, sufficiency and stability. Ablation tests are performed to remove certain elements from the proposed framework to determine the effect on overall performance. Five-fold cross-validation stratified on final outcome and grouped by student is used, so that no learner contributes to both training and test partitions, with paired t -tests over folds and Holm–Bonferroni correction [23] across the comparison family. Cohen’s d is reported alongside every p -value, since with large numbers of paired decisions significance is attainable at negligible effect size.

V. Results and Discussion

A. Predictive Accuracy Performance

The proposed framework was evaluated against four baselines: a majority-class predictor, SAKT, DKT and a GNN-based recommender. As shown in Table I, the suggested framework achieves the highest AUC-ROC at 0.9538 ± 0.0019 and the lowest RMSE at 0.2621 ± 0.0026 ,

exceeding DKT by 0.0103 ($p = 0.0009$) and the best baseline, the GNN-based model, by 0.0036 ($p = 0.022$) after Holm–Bonferroni correction. The results indicate that temporal knowledge tracing and graph-based concept links can be combined to get a more accurate representation of the changing knowledge state of the learner.

Two controls support that reading. The majority-class baseline returns exactly 0.5000, confirming the discrimination is not an artefact of the 67.9/32.1 class distribution, and SAKT — attention without graph conditioning or temporal decay — reaches 0.9233, so the improvement does not follow simply from using attention. Expected calibration error falls from 0.0084 to 0.0054 after temperature scaling; since predicted mastery is shown to learners inside explanations, a poorly calibrated probability is an incorrect statement rather than a cosmetic defect.

B. Learning Path Quality

The learning-path results show how successful sequential optimization is. Table II reports 906 recommendation decisions drawn from 400 held-out learners who passed or achieved distinction, with the concepts each learner actually studied next as ground truth. The proposed framework achieves the highest NDCG@5 at 0.1532 and the highest Hit@5 at 0.3687, ahead of weakest-first (0.1344), random selection among admissible actions (0.1300), popularity (0.1069) and syllabus order (0.0612). All differences are significant after correction.

The effect sizes qualify that conclusion, and we report them rather than the p -values alone. Against syllabus order the effect is small ($d = 0.36$); against random selection among admissible actions it is negligible ($d = 0.09$). The honest interpretation is that the prerequisite constraint performs most of the work and the learned ranking adds a small but consistent improvement on top of it. This also bounds the headroom available to a learned policy: an agent optimising the reward defined in Section III would have to improve on a greedy baseline that itself exceeds random selection by only $d = 0.09$.

Prerequisite violations are zero across all 906 decisions and every planner, so the guarantee of Section III is verified rather than assumed. Coverage of 0.726 shows the planner does not collapse onto a few popular concepts, the failure mode popularity-based recommendation exhibits here at 0.510.

C. Explanation Quality

The explainability module was evaluated by three objective measures rather than by participant rating, over 25 recommendation decisions each repeated under three perturbations. The sample is small and reported as such, but each measure is computed against the model’s own counterfactual response rather than against an opinion. Fidelity of 0.5318 indicates that cited prerequisite deficits are positively but imperfectly predictive of measured counterfactual improvement. Stability of 0.5310 indicates

that roughly half the cited gaps persist under a ± 0.02 perturbation of the mastery estimate.

Sufficiency of 0.0020 is the most informative of the three and is reported here as a negative finding. It shows that the concept named in an explanation accounted for only 0.2% of the predicted benefit, with the remainder arising from transfer to related concepts that the explanation did not mention. The explanations were accurate but substantially incomplete, and the renderer was subsequently amended to name the principal beneficiaries. No rating scale would have surfaced this, because participants cannot assess a quantity that the explanation omits.

D. Ablation Study

The lower rows of Table I report the conventional ablation. Removing knowledge-graph encoding changes predictive accuracy by -0.0001 ($p = 0.77$), and removing the temporal attention bias changes it by -0.0006 ($p = 0.45$). Read at face value, neither component contributes anything, which contradicts the expectation that knowledge-graph encoding supplies important structural information.

E. Why Conventional Ablation Is Insufficient

That reading is an artefact of the measurement rather than a property of the model. Predictive accuracy is computed only at assessment positions, and assessments exist for 89 of the 237 concepts, each carrying on the order of 2,700 labelled examples. A free embedding table learns such concepts adequately without any structural information, so the graph is genuinely redundant there. Its function is to support the remaining 148 concepts, which never enter the accuracy computation at all. The conventional ablation is structurally incapable of observing the component it removes.

To measure the contribution directly, all assessments were withheld for 27 of the 89 assessed concepts and their observed responses were removed from the model input, so that those concepts behave as genuinely unassessed; evaluation was then restricted to them alone. Table III reports the outcome: 0.9004 ± 0.0059 with graph propagation against 0.8679 ± 0.0042 without, a difference of $+0.0325$ ($p = 0.0003$, Cohen’s $d = 5.06$), with the graph-conditioned model superior on every fold.

Knowledge-graph propagation therefore contributes nothing where direct supervision is plentiful and substantially where it is absent, and a conventional ablation cannot distinguish the two cases. This has a practical consequence for the recommender, which must rank across all 237 concepts, most of which carry no assessment of their own; and a methodological consequence for the wider literature, in which component ablations are routinely reported on metrics that are unable to observe the component under test. We suggest that such ablations be accompanied by evidence that the reporting metric is capable of detecting the component being removed.

F. Overall Discussion

The experimental results demonstrate that combining explainability, knowledge-graph modeling, constrained path optimization and knowledge tracing improves both predictive and recommendation-oriented metrics relative to the assessed baselines. The explainability process enhances the interpretability of suggestions without being directly in charge of forecast accuracy. It does increase computational cost: the average time required to produce a fully explained recommendation is approximately 160 ms on CPU, against 7 ms for a cached mastery lookup alone. That trade-off indicates that the proposed framework provides substantially more transparency at an acceptable computational cost, and remains well within the latency budget of an interactive advising interface.

VI. Conclusion and Future Work

A. Conclusion

This study employs Knowledge Graph modeling, pedagogical explanation generation, prerequisite-constrained path optimization and Deep Knowledge Tracing to develop an Explainable AI based personalized route recommendation system for college students. The framework builds optimized learning paths by modelling learners’ changing knowledge states together with precondition links and temporal behavior, and the priority-backtracking process finds the knowledge gaps and dependencies that justify each suggestion.

On the Open University Learning Analytics Dataset the framework exceeded the majority-class, SAKT, DKT and GNN-based baselines, with an AUC-ROC of 0.9538 ± 0.0019 , an RMSE of 0.2621 ± 0.0026 and a calibration error of 0.0054. The planner produced zero prerequisite violations across 906 decisions evaluated against real learner behaviour, with the qualification that the prerequisite constraint rather than the learned ranking accounts for most of the margin over random selection.

The principal finding of the ablation study concerns evaluation. A conventional ablation indicates that knowledge-graph propagation contributes nothing, yet under a protocol that evaluates concepts lacking direct supervision the same component improves accuracy by $+0.0325$ with $d = 5.06$. The discrepancy is a property of the metric, not of the model: aggregate accuracy measured where supervision is abundant cannot observe a structural prior whose purpose is to compensate for its absence. The results indicate that learner-state modeling, structural knowledge representation, constrained sequence optimization and explanation generation together provide a transparent and personalized learning-path recommendation strategy for higher education — and that verifying such a claim requires evaluation protocols chosen to match what each component is for.

B. Limitations

The prediction target is successful completion rather than knowledge. Because OULAD records only submitted assessments, treating non-submission as failure is necessary for a usable class balance, but a substantial share of the signal is then engagement rather than understanding — appropriate for a system intended to prompt intervention, but a weaker claim than measuring mastery. The concept graph is likewise derived rather than given: OULAD supplies no concept annotations, prerequisite relations or material titles, so concepts are defined structurally and labelled by activity composition. Placement for 82% of materials relies on observed access, validated at $\rho = 0.975$ on the subset carrying both signals, but deriving concept order from behaviour and then mining prerequisites from behaviour is partially self-referential.

Supervision covers 89 of 237 concepts, so mastery for the remainder is not directly calibrated, and predicted mastery is bimodal, which is why learner-relative thresholds are required. The proposed model additionally conditions on learner features that the DKT and SAKT baselines do not, as those follow their original formulations; part of the reported improvement may therefore reflect feature access rather than architecture. Finally, all results derive from a single institution’s distance-learning data across seven modules, and generalisation to campus-based teaching is untested.

C. Ethical Considerations

All data used in this study is drawn from OULAD, which is publicly released under CC-BY 4.0 and contains no personally identifying information. No data was collected from human participants. The system produces advisory recommendations and makes no determinative decision concerning any learner. It does, however, condition on attributes including gender, disability status, index of multiple deprivation and age band, and a subgroup analysis across these attributes has not been conducted. Until one is, the system should not be deployed in any setting where its recommendations influence assessment or progression decisions.

D. Future Work

Four directions follow from the limitations above. The most immediate is a fairness analysis: subgroup performance across gender, disability, deprivation band and age band must be established before any deployment. Second, a learned policy over the state and reward of Section III should be trained and evaluated, in simulation and by off-policy estimation on logged trajectories; Section V quantifies the headroom, since the greedy planner exceeds random selection among admissible actions by only $d = 0.09$, and algorithms such as Soft Actor-Critic or Proximal Policy Optimization would have to widen that margin to justify their complexity. Third, a user evaluation would

TABLE I
Comparative Predictive Performance and Component Ablation

Model	AUC-ROC	RMSE	p
Majority (no-skill)	0.5000 ± 0.0000	0.4669	< 0.001
SAKT	0.9233 ± 0.0018	0.2992	< 0.001
DKT	0.9435 ± 0.0017	0.2751	0.0009
GNN-based	0.9502 ± 0.0014	0.2665	0.0220
Proposed	0.9538 ± 0.0019	0.2621	—
w/o knowledge graph	0.9539 ± 0.0021	0.2618	0.7704
w/o temporal attention	0.9532 ± 0.0014	0.2624	0.4507

Folds grouped by student; p Holm-corrected against the proposed model. Proposed ECE = 0.0054. Neither ablated component shows a detectable change — see Section V-E.

TABLE II
Learning Path Quality Comparison

Planner	NDCG@5	Hit@5	Viol.	Cov.
Syllabus order	0.0612	0.1832	0.0000	0.452
Popularity	0.1069	0.2936	0.0000	0.510
Random (admissible)	0.1300	0.3455	0.0000	0.722
Weakest-first	0.1344	0.3642	0.0000	0.515
Proposed	0.1532	0.3687	0.0000	0.726

906 decisions, 400 held-out successful learners; ground truth is the concepts each actually studied next. NDCG@5 vs. proposed: syllabus +0.0921 ($d = 0.36$); popularity +0.0464 ($d = 0.17$); random +0.0232 ($d = 0.09$); weakest-first +0.0189 ($d = 0.08$); all $p < 0.02$ Holm-corrected.

address what the objective measures cannot: fidelity, sufficiency and stability establish whether explanations are faithful to the model, not whether learners find them useful. Fourth, expert validation of the mined edges would give an independent estimate of graph precision, since the prerequisite relations are inferred from behaviour rather than supplied by a curriculum designer.

Beyond these, future studies can integrate multimodal information such as forum participation and video-lecture interaction, examine transfer across subjects and institutions, and create lightweight versions through pruning and distillation for deployment on resource-limited platforms.

Reproducibility

OULAD is available under CC-BY 4.0. All source code, SQL transformations and generated results are at github.com/abhay27singh/explainable-learning-path-recommender; every table here is regenerated from raw data by one command with fixed seeds.

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TABLE III
Cold-Concept Ablation

Variant	AUC-ROC	RMSE
With knowledge graph	0.9004 $\pm$ 0.0059	0.3359
Without	0.8679 $\pm$ 0.0042	0.3611
Difference	+ 0.0325 , $p = 0.0003$, $d = 5.06$	

All assessments withheld for 27 of the 89 assessed concepts, their responses stripped from the input, evaluation restricted to those concepts. Graph wins every fold.

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